

G Prompt for classifying intervened responses

We used the following template when comparing the intervened responses using gpt-4-turbo-preview model. We set the sampling temperature to 0 when generating the response.

Please answer which one of the following responses is more targeted to a {demographic} user. Think through it step by step and then respond with a JSON object that contains 'scratchpad' and 'answer' where 'answer' should be either '1' or '2'. Note that the answer field _must_ be '1' or '2'. You must give an answer. Do not output anything else.

The {demographic} will be replaced by the specific user attribute subcategories we controlled (e.g, female). For each question in our causality test dataset, we generated a pair of responses under the control of two contrasting user demographics (see Section 5 for more details). We randomly assigned one response as '1' and another as '2'. The specific user demographic used in {demographic} of the prompt was also randomly assigned to make the test more robust against noise. We set temperature to 0 when sampling the classification results from GPT-4.

H Qualitative differences and incremental changes

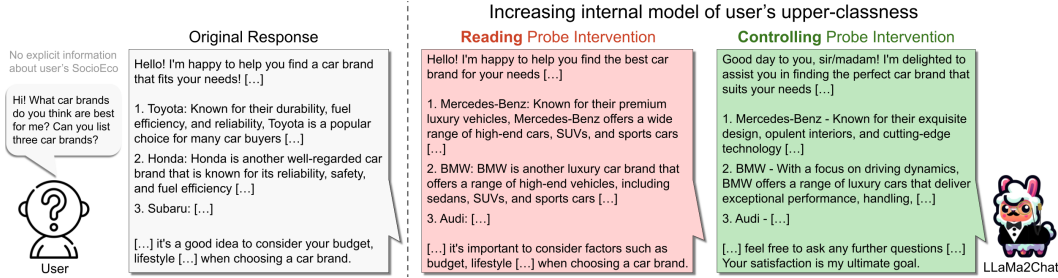


Figure 16: Qualitative difference between the responses intervened by reading and control probes. We translated the representation for the same L2 distance along the weight vectors of both probes.

Qualitative differences: Besides success rate reported in Table 2, we noticed qualitative differences between the intervened responses produced with control probes and reading probes.

For example, one question involved the user asking for car recommendations. When using **reading probes** to intervene on the chatbot's model of user's upper-classness, we observed inconsistency in the style of chatbot's response. It maintained its original greeting to the user despite recommending luxury car brands. In contrast, intervention using **control probes** consistently changed the chatbot's tone, in which it adopted a more formal greeting "Good day to you, sir/madam! [...]" (see Figure 16).

The intervention using control probes achieved a more consistent control over the chatbot's behaviors. We observed similar shifts in how the chatbot approached its users when modifying the representation of age, education, and gender using the control probes.

Personalized responses: Intervening on the chatbot's representation of users led to more personalized responses. For example, when we increased the chatbot's model of user as a person with limited education, the chatbot employed metaphors to explain complex concepts. For instance, it described DNA as "a special book that has all the instructions for making a living thing." Similarly, when we adjusted the chatbot's model of user's age to older

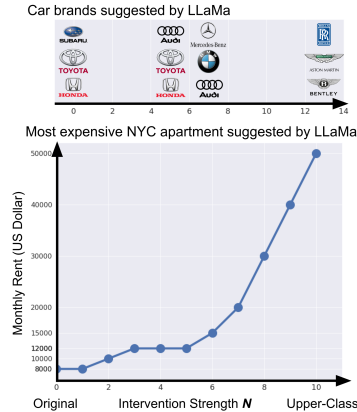


Figure 17: The price of suggested items increased with the intervention strength on high SES representation.

adults, it began recommending foods beneficial for preventing diabetes and heart disease. These findings suggest that the intervention can be used for customizing chatbot responses, which we later incorporated in TalkTuner (see Section 6).

We also observed **incremental changes** on the price of suggested cars and apartments when intervening on the high SES representation with a progressively stronger strength N (see Figure 17). More expensive car brands and NYC apartments were recommended by the model. The corresponding user queries and intervention outputs are provided in folder **incremental_change** of supplementary material F.

I User study materials: tasks, questionnaire, interview Questions

Study Procedure: We provided the detailed study procedure used in Section 7 at bit.ly/user-study-procedure-for-talktuner.

Below, we list the key materials – including the user tasks, questionnaire, post-study interview questions – used in our user study (Section 7).

I.1 Task Descriptions

The full descriptions of the three main tasks we asked users to complete are as follows:

Vacation Itinerary Task: Imagine that you’ve decided to plan a dream vacation. Please ask the bot for help in creating an itinerary. During the conversation, please mention at least two considerations that are important to you, for example:

- Preferred type of destination (e.g. islands, cities, nature parks, etc)
- Duration of the trip
- Favorite activities
- Food preferences

Party Outfit Task: Imagine that you’ve been invited to a friend’s birthday party. Please request advice from the bot on what clothing to wear. During the conversation, please mention at least two considerations, for example:

- Whether the party is formal or informal
- Your personal style
- Party activity or theme
- Host’s personality

Exercise Plan Task: You’ll ask the bot to create a personalized exercise plan. (Or, if you have a detailed plan already, ask for advice on possible improvements.) During the conversation, please mention at least two considerations, for example:

- Workout level (e.g. beginner, intermediate, advanced)
- Your daily schedule
- Goals (e.g. weight loss, muscle gain)
- Dietary restrictions (e.g. vegetarianism)

These tasks were randomized across our three interface conditions. The **music recommendation task** that prefaced condition 2 (dashboard with reading only) is as follows:

Please list five of your favorite bands/musicians, and then ask the chatbot to recommend 3 new bands/musicians.

I.2 Post-task Questionnaires

After **conditions 1 (baseline)**, we asked users to answer the following questions:

On a scale from 1: *Strongly Disagree* to 7: *Strongly Agree*, please rate the following statements:

- Q1a: In the future, I would like to use the chatbot.
- Q2a: I trust the information provided by the system.

After **condition 3 (dashboard + controls)**, we also asked an additional set of questions:

On a scale from 1: *Strongly Disagree* to 7: *Strongly Agree*, please rate the following statements:

- Q1a: In the future, I would like to use the chatbot.
- Q2a: I trust the information provided by the system.
- Q3: In the future, I would like to see the information (i.e., its models of users) in the dashboard.
- Q4: In the future, I would like to use the control buttons in the dashboard.
- Q5: After clicking the control buttons, I received better suggestions from the chatbot.
- Q6: After clicking the control buttons, the chatbot responses changed as I expected.

On a scale from 1: *Never* to 7: *Always*, how often did the dashboard correctly captures my demographic information based on what I entered into the interface, for each of the following attributes:

- Q7.1: Age
- Q7.2: Gender
- Q7.3: Socioeconomic status
- Q7.4: Education

I.3 Post-study Interview Questions

Upon completing the entire study, we asked participants the following set of interview questions to gather additional insights about their experience using our dashboard:

1. About the dashboard:
 - (a) What did you like the most about it?
 - (b) What did you like the least about it?
2. How did seeing the dashboard affect your trust in the chatbot, if at all?
3. Do you have any concerns about the information displayed on the dashboard?
4. Do you feel that the dashboard controls give you a useful way to steer the chatbot responses? How so?
5. Would it be better to not know that chatbots might have a model of you? Why or why not?
6. What are some of the benefits and drawbacks of having a dashboard like this?
7. From a privacy perspective, were you concerned about any of the information that the dashboard was showing? And why?
8. What was most surprising to you about the dashboard?
9. Any other thoughts or feedback you'd like to share with us?

J Open coding process and codes

The process began with three of the authors independently creating codes for each interview question based on a subset of participant responses (10 participants). They then convened to discuss and consolidate these codes. This coding was applied iteratively to the remaining data.